

What the schemes pay for: **every wheel written** — 8 bits means eight digits, 16 means sixteen, and the zeros in front are a mark · **working shown** — write the worths you added ($32 + 8 + 4 + 1$), not just the answer · **why binary** — a computer is built from switches that are only on or off · **the sentence** — "45 in denary is 00101101 in binary". Question 1 is a real Paper 1, verbatim; 2 to 7 are ours and say so. No computer needed for any of it.

1. March 2021 Paper 12 Q1(a)(b)(c), verbatim. A hockey club records the number of people that watch each match. An 8-bit binary register is used to store this value.

(a) 46 people watch the first match and 171 people watch the second match. Show how the registers would store these denary values as 8-bit binary. [2]

Denary value	8-bit binary
46	
171	

Working space

(b) Give the largest denary value that can be stored in the 8-bit binary register. [1]

(c) The hockey club wants to increase the number of people that can watch each match to 2000. The 8-bit binary register may no longer be able to store the value. Give the smallest number of bits that can be used to store the denary value 2000. [1]

2. Ours — and it says so. A counter has four wheels. Each wheel has two faces, 0 and 1. It shows 0101.

(a) Press +1 four times. Write the four numbers the counter shows, in order. [2]

(b) One of those presses turns a wheel that had never turned before. State what that wheel is worth. [1]

3. Ours — and it says so. The six cards from the lesson. Each card is one column of the count; the first number on a card is what its wheel is worth.

card ①	card ②	card ③	card ④	card ⑤	card ⑥
1 3 5 7 9 11 13 15 ...	2 3 6 7 10 11 14 15 ...	4 5 6 7 12 13 14 15 ...	8 9 10 11 12 13 14 15 ...	16 17 18 19 20 21 22 23 ...	32 33 34 35 36 37 38 39 ...

- (a) A friend's number is on cards ①, ③ and ⑥ and on no other card. State the number. [1]
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- (b) Write the number 45 as the list of cards it is on, then as 8-bit binary. [2]
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4. Ours — and it says so. Give the denary value of each binary number. Show the worths you added. [2]

- (a) 01010101 = _____ (b) 11101011 = _____
-
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5. Ours — and it says so. Convert each denary number into 8-bit binary. Show your working. [4]

- (a) 53 = _____
-
-
-
-
- (b) 170 = _____
-
-
-
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6. Ours — and it says so. A counter at a swimming pool's turnstile shows 0300 on its screen. The count is stored in a 16-bit register. Give the binary value stored in the register. [2]

7. Ours — and it says so. A computer is built from millions of switches. Each switch is either on or off. Explain why this means the computer uses binary to represent data. [2]



The ladder this week — Karel, not Python. There is no ice-cream challenge for this lesson, and that is deliberate: the code you write this week is **Karel puzzles 05 LightTrail and 06 TheLongHall** — your first loops. Binary is the paper half of the week; Karel is the keyboard half. Do both.



Dig, if you dare: build card ⑦. Which numbers are on it, what is its first number, and how high could you read minds with seven cards instead of six? Then say what card ⑧ would start with — without writing it out.



Above and beyond — not in the total. Binary is older than computers. **Leibniz wrote it down in 1703** — no electricity, no machines, just the observation that every number can be built from the wheels' worths. He also thought he had found it in the *I Ching*, a Chinese text roughly 3,000 years older than he was. Look up his paper's title, and then ask the honest question: *if binary needed no computer to exist, what did the computer actually add?*

Additional space. If you need more room for any answer, continue here and write the question number in the margin beside it.
